problem tags: physagent other writing problems writing need experiment expriment

General

We sincerely thank all the reviewers for their insightful and valuable comments!

We appreciate the reviewers' agreement that this paper presents a highly meaningful benchmark and dataset for **comprehensively assessing Vision-Language Models' understanding of the physical world** and exploring the **potential causes** of poor performance. Furthermore, we propose **a novel framework that enhances** physical reasoning capabilities. Our work also explores **the utility of our dataset and approach in robotic tasks**, contributing to the advancement of machine intelligence's understanding of the physical world, with promising implications for future research.

Overall, we are encouraged by the reviewers' feedback that:

- The motivation and novelty of PhysBench are clear, reasonable, and meaningful (all reviewers).
- The challenges faced by existing VLMs are well-articulated, and a well-defined, comprehensive benchmark is provided (all reviewers).
- The experiments are thorough, clearly demonstrating the gap in physical world understanding compared to common VQA benchmarks, identifying the reasons for VLMs' poor performance, and suggesting ways to improve their physical world understanding capabilities (all reviewers).
- The exploration of the causes of poor performance, scalability, and its impact on embodied AI is intriguing and has the potential to inspire future research (Reviewers 52XN, eQW6, khH4).

To address the concerns raised by the reviewers, we have conducted several additional experiments and analyses:

- We further validated the effectiveness of PhysAgent in three related benchmarks, as shown in Appendix ??? of the revised PDF.
- We added experiments evaluating some VLA models on virtual manipulation tasks, which have been synchronized and updated in Appendix ??? of the revised PDF.
- We have reorganized some sections to make the article clearer.

All changes are highlighted in *blue font*. We sincerely appreciate all the reviewers' suggestions and are committed to making further improvements.

Next, we address each reviewer's detailed concerns point by point. We sincerely thank all reviewers for their recognition of our work and the valuable suggestions provided. Discussions are always welcome. Thank you!

Reviewer 1 khH4 6

We sincerely appreciate your constructive and insightful comments. We will explain your concerns point by point.

> **Q1**: The test set of PhysBench is derived from the similar set of images as train. This ensures that any fine-tuning would result in performance improvement. Did the authors test PhysAgent against out of distribution images that are labeled for physical undersatnding?

physagent

A1: Thank you for your valuable suggestions! To demonstrate the effectiveness of our data and methods, we selected three benchmarks related to physical world understanding: ContPhy, which focuses on fluid dynamics understanding; Physion++, which emphasizes collision prediction; and Spatial VQA, which focuses on spatial understanding. Our findings indicate that...

Experimental results and details are provided in Appendix ?? of the PDF.

	ContPhy	Physion++	Spatial VQA
Phi-3V			
Phi-3V + finetune			
Phi-3V + PhysAgent			

> **Q2**: L443: 'PhysAgent first classifies the question and activaets task-specific prompts'. Does that mean that PhysAgent is dataset dependent? Also, how does it support the claim made at L90 "PhysAgent retrains the strong generalization abilities of VLMs and their capacity to solve open-ended problems"? It would be good to see an experiment supporting the L90 claim in the paper.

physagent

A2: Thank you for raising an important concern. We will address your question from the following three aspects:

(2.1) The classification of PhysAgent is automated. We have defined 20 distinct task types, with PhysAgent autonomously categorizing tasks based on their specific prompts. For example, if the task involves designing a light source, the corresponding module related to that task will be activated, using predefined instructions as mentioned earlier.

(2.2) To further demonstrate the generalizability of PhysAgent across a broader range of tasks, we selected three benchmark datasets related to physical world understanding for additional experimentation. The experimental results are presented in **Answer 1**.

> **Q3**: Is the PhysAgent using the DepthAnything, SAM and DINO outputs as input to LLMs or these individual model's outputs are used to get a textual insight which is the sole input to VLMs?

physagent

A3: We sincerely apologize for any unclear points in the paper. The three models are processed simultaneously and input into the LLMs; however, whether the results from these visual foundation models are considered in the decision-making process is determined by the LLMs themselves. This design facilitates the integration of more advanced visual perception capabilities, addressing a key error identified in Section 3.4.

We commit to making the appendix more concise and adding clarifications to improve the readability of the text. We have already made some revisions to the original manuscript and will continue to refine it to enhance its readability.

We hope the above discussion resolves your concerns. The discussion remains open, and we welcome your further review.

Reviewer 2 52XN 5

Overall, we are encouraged by the recognition and deeply appreciate your valuable suggestions. Moving forward, we will address each of your concerns point by point.

> **Q1:** I am a bit confused about the correlation matrix in Figure 4a. I am assuming the last 4 rows are PhysBench splits? In that case, the bottom right half seems to indicate that that performance on their benchmark doesn't correlate on the other existing VLM benchmarks (since the left part of the bottom 4 rows are blue). This can imply improving on their benchmark doesn't improve on other vision-language benchmarks? I am also not sure what part of that figure to look at for their claim in Section 3,4 that their benchmark is correlated to MMMU.

writing

A1: Thank you very much for your valuable question! We apologize for any unclear aspects in the figures and the paper.

(1.1) Your understanding is correct. The last 4 rows represent the PhysBench splits. As you can observe, PhysBench has relatively low correlation with conventional VQA benchmarks, further highlighting the distinctiveness and uniqueness of our task compared to traditional tasks.

(1.2) The large differences between individual tasks do not directly lead to the conclusion that "improving on their benchmark doesn't improve on other vision-language benchmarks." However, as demonstrated in the experiments in Section 3.4, expanding model size and data scale leads to significant improvements for conventional VQA tasks, but this does not hold for PhysBench. This discrepancy may be due to the lack of physical world-related knowledge in the data. Additionally, for tasks like Pope, which involve hallucination detection, improvements are not always observed with increases in data and model size. This finding resembles the behavior seen in PhysBench, and we can also observe their correlation in Figure 4a.

(1.3) Figure 4a shows that the task most similar to MMMU is SQA, followed by the "property" and "relationship" subcategories of PhysBench.

Overall, Figure 4a's exploration of relationships between different tasks partially reveals the uniqueness of PhysBench compared to conventional VQA tasks.

> **Q2:** More details on the PhysAgent would have been helpful. What is the base model for PhysAgent - is it an MLM? If so, which one? They outline a 3-step pipeline:
>
> 1. asking task-specific prompts: How does it learn to ask these questions? Is it tuned to do so or in-context? If so, how is the tuning done?
> 2. Incorporating VLM predictions: How exactly do they incorporate depth, grounding etc? The figure 8 and supplementary makes it seem like a JSON-like format or a list. Where does the knowledge memory come from? Can the PhysAgent interpret such a format without any fine-tuning?
> 3. How is PhysAgent prompted to do COT? Does it do it zero-shot or, is it taught to do the chain of thought using some training examples? If so, how are the COT training examples made?

physagent

A2: Thank you for your suggestion. PhysAgent is centered around a Vision-Language Model (VLM), and it can utilize any VLM. In our paper, we conducted experiments using the open-source Phi-3V and GPT-4o models.

(2.1) We predefined 20 task categories and required the VLMs to classify input questions and activate corresponding physical world knowledge to assist with in-context question answering. PhysAgent and tuning are not contradictory; PhysAgent essentially builds a series of processes based on a VLM at its core, making full use of prior physical knowledge and the perceptual capabilities of powerful foundational vision models. The lack of these two factors is the primary reason for poor performance in VLMs. Additionally, the experiments in our paper typically involved either tuning or zero-shot usage of PhysAgent, without combining both approaches.

(2.2) The three models are processed simultaneously and input into the LLMs; however, whether the results from these visual foundation models are incorporated into the decision-making process is determined by the LLMs themselves. Knowledge memory is predefined by us and consists of two parts: one part contains general properties of objects, such as the high ductility of gold, while the other part includes physical inferences, such as the direction of shadow movement relative to the light source.

(2.3) The CoT in PhysAgent is zero-shot. PhysAgent uses CoT in two parts: when providing an answer, PhysAgent is required to explain it step by step. After the answer is given, we prompt PhysAgent a second time to verify the reasonableness of its own answer.

> **Q3:** The results in Table 9 are confusing - what the individual columns? They do not match the names with the splits in PhysBench in Table 3. PhysAgent improves on PhysBench, but does it at least retain performance on some other similar benchmarks?

writing

A3: Thank you for your suggestion! We apologize for any unclear aspects in the paper.

(3.1) The details for Figure 9(a) are as follows, and we have included additional information in the revised version's Appendix ???. The results are as shown. (We noticed that Table 9 is the hyperparameter table, and we believe you may be referring to Figure 9(a). Please let us know if you have any further questions.)

	number	mass	color	attribute	size	location	depth	distance	movement	temperature	camera	gas	light	collision	throwing	manipulation	fluid	chemistry	others
Phi-3V	43.15	37.88	63.00	40.81	66.07	53.61	74.39	48.33	26.69	52.94	38.93	65.45	29.48	33.18	35.39	28.95	35.34	59.46	56.68
+ Desp-CoT	49.22	29.01	67	37.62	66.07	53.61	62.46	55	24.69	58.82	40.15	67.27	30.3	34.99	32.07	28.47	31.94	56.76	50.68
+ PLR	43.67	26.62	59.67	34.63	58.93	41.92	30.53	43.33	30.43	67.65	39.96	67.27	30.94	32.13	28.03	25.69	35.03	56.76	51.5
+ PhysAgent	46.27	37.88	64.67	40.53	67.86	60.82	69.47	66.67	26.76	72.06	38.93	70.91	34.58	36.05	35.39	29.55	35.49	67.57	65.67
GPT-4o	61.18	54.27	71.00	52.52	85.71	70.45	73.33	83.33	60.58	91.18	22.05	83.64	32.67	43.74	46.32	35.22	39.20	62.16	86.92
+ CoT	63.43	56.66	71.67	54.22	82.14	74.23	75.44	83.33	67.91	91.18	30.73	74.55	36.49	45.40	46.32	35.71	41.67	64.86	85.56
+ Desp-CoT	65.16	55.29	71	52.16	80.36	68.38	70.18	71.67	61.2	85.29	29.41	80	39.95	44.19	42.28	30.52	43.83	70.27	85.83
+ PLR	68.63	41.98	66.00	47.27	66.07	57.73	59.3	53.33	43.91	88.24	30.73	89.09	34.94	38.01	34.68	24.85	37.5	66.22	79.56
+ PhysAgent	67.07	53.58	73.33	55.86	87.50	79.04	59.3	95.00	71.99	95.59	41.19	85.45	43.49	45.70	46.32	52.11	44.91	67.57	87.47

(3.2) To demonstrate the effectiveness of our data and methods, we selected three benchmarks related to physical world understanding: ContPhy, which focuses on fluid dynamics understanding; Physion++, which emphasizes collision prediction; and Spatial VQA, which targets spatial reasoning. Our findings indicate that...

Experimental results and details are provided in Appendix ?? of the PDF.

	ContPhy	Physion++	Spatial VQA
Phi-3V			
Phi-3V + finetune			
Phi-3V + PhysAgent			

We hope this clarifies your concerns. We are committed to thoroughly incorporating your suggestions in the next version of the paper. Thank you once again for your excellent feedback.

Reviewer 3 xxa5 5

We sincerely thank you for your comprehensive comments and constructive advice. We will explain your concern as follows.

> **Q1:** Discussion of Related Work: The paper lacks a comprehensive discussion of related works. The proposed benchmark aims to evaluate and enhance the ability of VLMs to understand the physical world at both the object and scene levels. However, it is crucial to emphasize that understanding 3D spatial relationships at the scene level is a fundamental capability for VLMs. To strengthen the discussion, the authors should compare this work with other 3D spatial VQA benchmarks such as SQA3D [1], OpenEQA [2], EmbodiedScan [3], MMScan [4], and MSQA [5]. These benchmarks offer direct evaluations of VLMs' understanding of 3D scenes through interleaved 2D or 3D representations, providing valuable context and highlighting gaps or overlaps with the proposed approach.

writing

A1: Thank you for your insightful question. We conducted additional ablation experiment

Thank you for your valuable suggestions! In fact, we discuss Spatial VQA datasets in L117, which emphasize geometric relationships, representing only a part of physical world understanding. We focus on a more comprehensive evaluation of physical world perception capabilities. However, the works you have mentioned are also highly valuable to us. We have further discussed these references in Appendix ??? and added the appropriate citations.

> **Q2:** Comparison of Embodied Tasks: The comparison between zero-shot models (such as existing VLMs and PhysAgent) and fine-tuned models trained on PhysBench raises concerns about fairness. PhysBench, in its current form, is relatively simplistic and may not provide a robust or balanced evaluation. To ensure fair comparisons, the authors should either (1) incorporate more challenging tasks or datasets in PhysBench to match the capabilities of fine-tuned models, or (2) highlight the limitations of comparing zero-shot performance with fine-tuned results to avoid misleading conclusions.

A2: Thank you for your constructive suggestions. Following your recommendations, we have conducted additional experiments on 5 benchmarks across 3 task types to further evaluate the effectiveness of our method in **Table E**.

writing

I don't understand

> **Q3:** Recently, several open-source VLA models have emerged, such as OpenVLA [1], Embodied-COT [2], and Octo [3]. Did the authors consider comparing the fine-tuned VLMs with other VLA models on the proposed manipulation tasks? Additionally, is there a way to assess the benchmark's role in advancing the development of VLA models? Such comparisons and evaluations would further highlight the benchmark's contribution to the field.

A3: Thank you for your question. We apologize if our manuscript gave the impression that the training recipe for Sugar was unclear.

expriment

Thank you very much for your insightful comments! Currently, there are two approaches for applying VLMs to robotic manipulation: (a) directly adjusting VLMs to VLA or training VLA models (e.g., OpenVLA), and (b) using VLMs as agents. Our experiments are based on the agent-based MOKA framework. Unfortunately, we are unable to validate the effectiveness of our data or method through simple fine-tuning or in-context approaches on models using the (a) approach, such as OpenVLA. Nevertheless, we still conducted experiments:

	Affordance	Force	Color	Location	Tool
MOKA					
MOKA + finetune					
MOKA + PhysAgent					
Octo					
OpenVLA					

Both OpenVLA and Octo require fine-tuning to achieve better performance, as they need to perform action de-regularization to obtain the correct action scale (a process dependent on data training). On the other hand, the execution in MOKA and similar frameworks depends on the 2D-to-3D action trace, which is data-independent. Therefore, MOKA performs better on our benchmark.

(1) On one hand, based on the fact that a better VLM can enhance VLA performance [1], our data can provide VLMs with stronger physical world perception capabilities, which may also benefit VLA. (2) On the other hand, incorporating VQA tasks during VLA training can help improve the model's generalization [2], and we believe our data could provide potential assistance.

As mentioned earlier, due to space limitations, we are currently focusing on the second paradigm and have demonstrated the usefulness of our data. The discussion of the first paradigm will be reserved for future work.

We commit to including the above discussions in the relevant sections of the appendix in future papers.

[1] TinyVLA: Towards Fast, Data-Efficient Vision-Language-Action Models for Robotic Manipulation
[2] HAMSTER: Hierarchical Action Models for Open-World Robot Manipulation

Thank you once again for your recognition and constructive suggestions, which have been instrumental in enhancing the quality of our research!

Reviewer 4 eQW6 6

We sincerely thank you for the valuable comments and we will explain your concern as follows.

> **Q1:** The article is quite lengthy, so it is really important to improve its logical flow to enhance readability. Using appropriate formatting elements, such as bold text for key points, can help guide readers through the content and make it easier for them to locate the information they are looking for. It is difficult for me to find my questions when reading this paper.

writing

A1: We are committed to making the appendix more concise and providing additional clarifications to improve the text's readability. We have already implemented some revisions to the original manuscript and will continue refining it to enhance its overall clarity and flow.

> **Q2:** There is a lack of consistency in the article's formatting. For example, task names are referred to in multiple ways, and similar tables or images are styled differently across various sections, which may confuse readers. Specifically, Table 9a labels the four main types as Property/Spatial/Envir./Phe., whereas other sections refer to them as Property/Relations/Scene/Dynamics. Are they describing the same thing? Also, styles for Tab 16-20 are different, and what is the last column of Tab 20? The labels above Figure 4 are misaligned with the ticks, and extra ticks appear at the bottom. I believe these ticks should likely be placed on the left side instead. It is better to re-organize the whole paper.

writing

A2: Thank you for your suggestion! We apologize for any unclear or ambiguous aspects in the paper. We have standardized the task names and unified the content of Tables 16-20, as well as the last column, which have been updated in the revised version of the PDF. Once again, we sincerely appreciate your thoughtful feedback!

> **Q3:** How do questions design for each video? The paper mentions, "For videos annotated with physical principles, we generate physics-related questions using both manual design and GPT-4o, following predefined rules.". But I cannot find further details on it. What are the question templates? How do humans design questions? What is the procedure of question generation by GPT-4o? The authors may comment on more details on this matter.

writing

todo

A3: Thank you for your insightful query!

> **Q4:** How is the human evaluation in Tab 3 (or other possible places) conducted? Also, how do human annotators contribute to the dataset? I can only find the GUI description for this matter. The authors may comment on more details, such as how many people are engaging in each part and so on.

writing

A4: We apologize if our manuscript suggested that how do human annotators contribute to the dataset was unclear.

to

> **Q5:** What is the detailed performance on 19 tasks among PhysAgent and baselines of Fig 9a, since the distribution of the dataset is not balanced.

writing

A5: Thank you for your insightful question.

The details for Figure 9(a) are as follows, with additional information provided in the revised version's Appendix ???. The results are presented as follows.

	number	mass	color	attribute	size	location	depth	distance	movement	temperature	camera	gas	light	collision	throwing	manipulation	fluid	chemistry	others
Phi-3V	43.15	37.88	63.00	40.81	66.07	53.61	74.39	48.33	26.69	52.94	38.93	65.45	29.48	33.18	35.39	28.95	35.34	59.46	56.68
+ Desp-CoT	49.22	29.01	67	37.62	66.07	53.61	62.46	55	24.69	58.82	40.15	67.27	30.3	34.99	32.07	28.47	31.94	56.76	50.68
+ PLR	43.67	26.62	59.67	34.63	58.93	41.92	30.53	43.33	30.43	67.65	39.96	67.27	30.94	32.13	28.03	25.69	35.03	56.76	51.5
+ PhysAgent	46.27	37.88	64.67	40.53	67.86	60.82	69.47	66.67	26.76	72.06	38.93	70.91	34.58	36.05	35.39	29.55	35.49	67.57	65.67
GPT-4o	61.18	54.27	71.00	52.52	85.71	70.45	73.33	83.33	60.58	91.18	22.05	83.64	32.67	43.74	46.32	35.22	39.20	62.16	86.92
+ CoT	63.43	56.66	71.67	54.22	82.14	74.23	75.44	83.33	67.91	91.18	30.73	74.55	36.49	45.40	46.32	35.71	41.67	64.86	85.56
+ Desp-CoT	65.16	55.29	71	52.16	80.36	68.38	70.18	71.67	61.2	85.29	29.41	80	39.95	44.19	42.28	30.52	43.83	70.27	85.83
+ PLR	68.63	41.98	66.00	47.27	66.07	57.73	59.3	53.33	43.91	88.24	30.73	89.09	34.94	38.01	34.68	24.85	37.5	66.22	79.56
+ PhysAgent	67.07	53.58	73.33	55.86	87.50	79.04	59.3	95.00	71.99	95.59	41.19	85.45	43.49	45.70	46.32	52.11	44.91	67.57	87.47

> **Q6:** Are there any reasoning steps included in each question?

writing

A6: To simplify our setup, we directly used an option-based format to respond, as illustrated in the example in Figure X of the paper. During our design process, we found that evaluating the correctness of the reasoning process is highly challenging [1]. Additionally, if answers are given in an open-ended format, using LLMs for evaluation introduces a series of other issues [2]. In our experiments in Section 3.4, we also found that when using Chain-of-Thought (CoT), the error rate for answer extraction by GPT was as high as 2%.

However, this does not imply that our work overlooks the importance of reasoning steps. In Section 3.4, we manually reviewed the reasoning processes of three representative models, summarized the types of errors, and observed that while reasoning errors are significant, a larger issue lies in the lack of relevant physical world knowledge and perception errors. As a result, we further propose PhysBench to enhance physical world perception.

[1] The challenge of evaluating the correctness of the reasoning process (need reference paper)
[2] Using LLMs for evaluation also raises a series of other issues (need reference paper)

Thank you for your valuable comments. We have revised our manuscript based on your suggestions and are committed to further refining the paper to improve its readability. We sincerely appreciate your help once again!